

**Effects on Maternal Bandwidth of Monthly Unconditional Cash Starting at Birth
Among U.S. Families with Low Income**

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Abstract

Financial scarcity has been posited to reduce adults' cognitive bandwidth. Evidence of the causal impact of reducing poverty and improving cognitive bandwidth is thin, particularly in the context of the United States and for the parents of young children. We examine the impact of a monthly unconditional cash transfer on cognitive bandwidth of U.S. mothers with low income ($N=828$). Preregistered intent-to-treat analyses find no differences in cognitive bandwidth of mothers after receiving a high-cash gift (\$333/month) versus a low-cash gift (\$20/month) for 48 months after childbirth.

Keywords: randomized controlled trial, cognitive bandwidth, cash transfer, poverty

Introduction

Recent research has posited that poverty and financial instability can overburden cognitive resources and thereby reduce cognitive bandwidth [1,2]. Cognitive bandwidth is the cognitive capacity and attention required for effortful thought, decision-making, and follow-through (Mullainathan & Shafir, 2013, p. 13), and this capacity is essential for navigating everyday life [3]. It has been theorized that scarcity may reduce cognitive bandwidth by causing attentional focus on scarcity-related demands and attentional neglect to other demands [4,5]. People experiencing poverty are often facing a variety of issues that demand their attention, which can leave less cognitive capacity available for new tasks that arise.

However, the empirical relationship between financial scarcity and cognitive bandwidth is unclear, particularly in the context of adults with low income in high-income countries such as the United States. [6]. Some quasi-experimental work has found associations between financial scarcity and reduced cognitive bandwidth, primarily in the context of low- and middle-income countries [4,7–9]. More specifically, Mani et al. (2013) found diminished cognitive performance among Indian sugarcane farmers before harvest (when financial scarcity is high) compared with after harvest (when financial scarcity is low). Additionally, Ong et al. (2019) showed that a large, unanticipated debt-relief program in Singapore improved cognitive functioning among adult men and women. A recent randomized controlled trial in Finland found that a monthly unconditional cash transfer (partial basic income of €560/month) for 2 years was associated with increased ability to concentrate in a sample of unemployed adults [10].

In contrast, other studies have reported no association between financial scarcity and adult cognitive bandwidth [11,12]. Szaszi et al. (2023) found that a one-time unconditional cash transfer to low-income men in Liberia did not affect performance on two executive function

tasks measuring attention, inhibition, and working memory. González-Arango et al. (2022) could not replicate Mani et al.'s (2013) findings in a study of high school students in urban Colombia. In the U.S. context, Carvalho et al. (2016) randomly assigned low-income adults to complete a survey either before or after an earnings payday and found no differences in performance on inhibitory control or working memory tasks.

One potential explanation for the mixed results in the previous literature may be the varied ways cognitive bandwidth has been conceptualized and measured. Most studies have utilized measures of executive functions – such as selective attention and inhibitory control (such as the Flanker task) and/or working memory (such as a digit span task) – as measures of cognitive bandwidth. However, other studies have assessed general nonverbal cognitive ability (such as Raven's matrices) or have even utilized self-reports of recent ability to concentrate. Though no clear patterns have emerged by cognitive bandwidth measure thus far, it is possible that effects of scarcity on cognitive bandwidth are sensitive to these conceptual and methodological differences. Additionally, while participants in previous research studies share the context of having low income or residing in poverty, they differ in other characteristics including gender, age, employment, and whether or not they are parents. Such participant characteristics may play an important role when considering the association between financial scarcity and cognitive bandwidth.

Financial Scarcity and Parenting

To date, the literature on financial scarcity and cognitive bandwidth has not focused on the context of parenting. However, this relationship may be particularly important for parents because the potential impact of financial scarcity on cognitive bandwidth may have spillover and feedback effects on parenting behaviors, parental time, and general quality of parenting [14–18].

For parents with low income, financial scarcity and uncertainty might affect availability and allocation of attention towards parenting, whether directed to meeting daily needs, spending time in nurturing interactions with children, or addressing a range of other parenting demands [1,4,19]. Limited or redirected attention and related cognitive resources may be implicated in differences in the quality of parenting behavior across socioeconomic groups [20].

Despite the extensive literature linking poverty and child development outcomes [21], there is very little empirical evidence on the extent to which family economic resources *causally* determine early life development and we are aware of no empirical evidence on the extent to which low economic resources affect early development through channels of parental cognitive resources and parental investments.

Financial Scarcity and Income Variation Throughout the Month

Financial scarcity can be both chronic or intermittent, including among very low-income individuals and families, who may have less access to economic cushions such as savings or insurance [1,22]. Income instability or intermittence can be experienced over different temporal periods [23]. Individuals or families with very low income may experience more or less financial scarcity depending on the timing of receipt of earnings paychecks or social benefits [24]. Indeed, within-month volatility in income receipt among low-income individuals and families is correlated with timing of spending (on dining, gas, stores, and online purchases; Jacobe & Jones, 2009), nutritional intake [26], and children's school behavior and performance [27–29]. To date, there is no empirical work investigating how timing of income receipt within a month co-varies with maternal cognitive bandwidth in the U.S. context.

98 **The Present Study**

99 The empirical link between poverty and cognitive bandwidth—particularly in the U.S.
100 context and among parents—remains unclear. Specifically, little is understood about the
101 potential impact of reducing financial scarcity, and the incremental benefits of stability and
102 predictability, on cognitive bandwidth.

103 Here we investigate whether an unconditional cash transfer affects mothers' cognitive
104 bandwidth by leveraging the Baby's First Years (BFY) study to assess the causal impact of four
105 years of monthly unconditional cash transfers among a large, racially and ethnically diverse
106 sample of U.S. mothers with low income. The prior conceptual and empirical literature posits
107 that predictable, stable monthly income will reduce poverty and increase financial stability, in
108 turn enhancing parental attention and self-control. Based on this literature, we preregistered our
109 hypothesis that mothers receiving a higher monthly cash gift will have higher performance on
110 (an objective assessment of) cognitive bandwidth than mothers receiving a lower monthly cash
111 gift.

112 Second, we add to a broader literature regarding timing of income receipt and the impact
113 on outcomes of within-month financial scarcity by exploring how timing of the monthly cash gift
114 affects maternal cognitive bandwidth. The cash gift is disbursed monthly on the day of the
115 child's birth date; it is predictable and exogenous to other sources of income. Analyses of
116 transactions on the debit card on which the cash gift money is disbursed show that the higher
117 cash gift money is typically drained each month and used within days of each monthly
118 disbursement [30,31]. Thus, the cash gifts offer an empirical opportunity to examine how timing
119 of the cash gift receipt relative to probable experiences of financial scarcity within a month co-
120 varies with maternal bandwidth. In these exploratory analyses, we hypothesized that maternal

cognitive bandwidth would be higher after the monthly cash gift disbursement and lower as the time between cash gift disbursement and assessment increases.

The present study makes several important contributions to the literature. Firstly, we examine the causal impact of predictable monthly unconditional cash on cognitive bandwidth via a randomized controlled study, as a test of the scarcity hypothesis posited by Mullainathan & Shafir (2013). Secondly, we examine this question among adults with low income in the context of a high-income country (the United States). Our third main contribution is the examination of this question in a sample of mothers of young children specifically. The first several years of a child's life are a particularly sensitive period of child development in which parenting demands are especially high. Reduced bandwidth may affect parent investments and parenting quality, which are valuable inputs for supporting child development.

Method

We leverage the ongoing BFY study (Noble et al., 2021), a randomized controlled trial providing monthly unconditional cash gifts to mothers with low income in the United States starting at childbirth for the first several years of their child's life.

Primary impact analyses were preregistered on [clinicaltrials.gov](https://www.clinicaltrials.gov) (<https://www.clinicaltrials.gov/study/NCT03593356>), the Registry of Efficacy and Effectiveness Studies (Registry ID: 1405.1v6), and the AEA RCT Registry (<https://www.socialscienceregistry.org/trials/3262>). All materials used in this study (including full maternal surveys at each wave) are publicly available at <https://www.babysfirstyears.com/data-and-documentation>. Anonymized data used in this manuscript will be made available to researchers via ICPSR

(<https://www.icpsr.umich.edu/web/DSDR/studies/37871/> [33]), and analysis code will be available via openICPSR (<https://www.openicpsr.org/openicpsr/project/159422/>).

Study Design

From 2018 to 2019, the BFY randomized controlled trial recruited 1,000 mothers with low income across 12 hospital maternity wards in four U.S. cities (New Orleans, LA; New York, NY; Omaha, NE; Twin Cities, MN). Mothers were approached about participating in a longitudinal child development study shortly after giving birth; inclusion criteria included the child being the product of a singleton pregnancy, admitted to the well-baby nursery, and going home with the mother; and the mother being of legal age for informed consent, speaking English or Spanish, having the intent to reside within 50 miles of the birth hospital, and self-reporting an income at or below the federal poverty line.

Mothers who subsequently agreed to the opportunity to receive a monthly cash gift were randomized within each site to receive \$333 (“high-cash gift”; 40% of $N=1,000$) or \$20 (“low-cash gift”; 60% of $N=1,000$) for the first several years of their child’s life. The cash gifts were disbursed onto a debit card (co-branded “4MyBaby”) monthly on the day of the child’s birth date [34]. This variation in disbursement date throughout the month therefore is independent from receipt of other sources of income. Study follow-up response rates are very high (90%, $n=890$ at age 4; see Supporting Information, Fig A.1 for CONSORT diagram), and the mechanism of cash gift disbursements worked well, with very high uptake and few problems: nearly all families have used the BFY cash money on the debit card at least once [31].

Assuming 20% attrition, a sample size of $n=800$ mothers at the wave 4 follow-up, when the focal child is age 4 years, gives 80% statistical power to detect an impact of .207 standard deviations at $p<.05$ in a two-tailed test.

Measures of Maternal Cognitive Bandwidth

Mothers and their children participated in an in-person visit at a local university around the time of the child's 4th birthday. Baseline demographic characteristics for participants included in the present analyses were balanced between the low- and high-cash gift groups; see Supporting Information, Table A.1.

During the visit, mothers completed a survey and a direct assessment of cognitive bandwidth measured via the Minnesota Executive Function Scale (MEFS; Carlson & Zelazo, 2014). The MEFS is a tablet-based adaptation of a dimensional change card sort task, which is a commonly used executive function assessment for children and adults. The MEFS is reliable and valid [36] and has several strengths compared with other measures, including those used in prior studies (e.g., Raven's Progressive Matrices in Schofield & Venkataramani, 2021; numeric Stroop in Mani et al., 2013). For example, Raven's Progressive Matrices measures general intellectual ability and abstract reasoning, while the MEFS is an assessment of executive function skills (working memory, cognitive flexibility, and inhibitory control), and is not strongly correlated with IQ [36]. Additionally, unlike the numeric Stroop task, the MEFS has been normed via a large racially diverse sample of adults across the United States [36], enabling comparison of BFY mothers' scores to national norms. We use MEFS standard scores (standardized to $M=100$, $SD=15$, referencing the national norms).

The BFY study sample for examining impacts on MEFS consists of 357 mothers in the high-cash gift group and 471 mothers in the low-cash gift group. An additional 12 mothers are excluded from these analyses for the following reasons: experimenter error, $n=4$; technical error, $n=4$; child/sibling fussiness, $n=3$; refusal to participate, $n=1$. (Of these 12 mothers, 9 were in the low-cash gift group and 3 were in the high-cash gift group; 5 were in Omaha, NE; 3 in New

York, NY; 2 in Twin Cities, MN; and 2 in New Orleans, LA.) Baseline characteristics for the final sample, listed in Table A.1, are balanced (joint test: $p=.505$).

Analyses of MEFS are supplemented with a self-reported measure of maternal attention using an item from the Patient Health Questionnaire (PHQ-9; Kroenke & Spitzer, 2002), which was administered in a maternal survey during each of the four follow-up waves of data collection (at approximately child ages 1, 2, 3, and 4 years). This item asks mothers to rate how often they have “Trouble concentrating on things, such as reading the newspaper or watching television” on a scale from 1 (“Not at all”) to 4 (“Nearly every day”). We use two operationalizations: continuous (1–4) as well as dichotomous (scores of 1, indicating no trouble concentrating, vs. scores 2–4, indicating any trouble concentrating; as in Kangas et al., 2019).

Empirical Approach

First, we estimate the causal impact of the high-cash gift on maternal MEFS scores via preregistered intent-to-treat regression analyses. We conduct this analysis with and without covariates: site, child age at interview in months, and a set of baseline characteristics (mother’s age, completed schooling, household income, net worth, general health, mental health, race and ethnicity, marital status, number of adults in the household, number of other children born to the mother, smoked during pregnancy, drank alcohol during pregnancy, father living with the mother, child’s sex, birth weight, and gestational age at birth). We calculate intent-to-treat estimates using simple linear regression:

$$(1) Y_{is} = Z_{is}\pi + X_{is}\beta + \delta_s + \varepsilon_{is}$$

where Y is the outcome measure for each individual i in site s , Z is an indicator for cash gift group assignment, X represents the baseline covariates as well as the child’s age in months, π

captures the impact of the cash gift, δ_s is a fixed effect for each site (New Orleans, LA as the omitted site) and ε is the error term.

We estimate the causal impact of the high-cash gift on the PHQ-9 concentration item using the same approach at each wave of data collection (child ages 1–4) as well as pooled across the four waves as depicted here:

$$(2) Y_{iws} = Z\pi_{iws} + X_{iws}\beta + \delta_s + \omega_w + \varepsilon_{iws}$$

The regression-adjusted model (2) includes an indicator for whether the follow-up survey was conducted in person or by phone in the first wave (to capture the onset of the COVID-19 pandemic during fielding in X), an indicator for each wave of follow-up (ω for each wave w), and the child's age in months relative to the time of the survey interview in X .

Next, we explore whether the timing of the MEFS administration to mothers within a month was associated with maternal MEFS scores. Equation (1) is expanded as follows:

$$(3) Y_{iws} = Z\pi_{iws} + X_{iws}\beta + T_t\beta_1 + T_tZ\pi_{2iws} + \delta_s + \mu_m + \varepsilon_{iws}$$

Here, T represents indicators capturing dimensions of temporal variation ($t \pm x$) within a month or with respect to the timing of the receipt of the cash gift, for example, a matrix of weeks of the month would translate to T_1 (calendar week 2), T_2 (calendar week 3) or T_3 (calendar week 4), or a matrix of days since the previous cash gift ($T_{1=2}$ weeks since cash gift; $T_{2=3}$ weeks since cash gift, etc). We consider the week of the month (calendar weeks 1–4), weeks since previous cash gift disbursement (1–4), day of the month (calendar date 0–31), and number of days since the previous cash gift disbursement (0–31). These analyses control for month of assessment μ (January–December, omitting August), site, language of assessment (English or Spanish), and the previously described full set of covariates collected at baseline.

Results

Descriptive Findings

Standardized MEFS scores among BFY mothers (97.60; $SD=14.70$) are near the population mean and standard deviation of 100 ($SD=15$), based on the MEFS norming sample. This suggests that the scores of mothers in the BFY sample are in line with U.S. population norms, despite those norms being derived from a more socioeconomically diverse population. BFY maternal MEFS scores are negatively correlated with maternal age ($r=-0.15$, $p<.001$) and positively correlated with years of education ($R=0.12$, $p<.001$; Fig 1). On average, mothers with less than a high school degree scored 93.17 ($SD=12.31$), while mothers with a high school degree scored 97.96 ($SD=15.71$) and mothers with more than a high school degree scored 101.57 ($SD=15.08$).

Fig 1. Associations between mothers' MEFS scores and demographic variables. Outliers ($>3SD$ from mean) are excluded. R and p -values correspond to zero-order Pearson's correlations.

Correlations with household income in the low-income BFY sample are mixed. For these analyses, we excluded any data point that was more than three standard deviations from the mean. BFY mothers' MEFS scores are correlated with total household income measured at the fourth wave of follow-up (i.e., representing income in the year prior to the measurement of MEFS, calendar year 2021–2022), but not with household income measured at study entry ($R_{\text{age4}}=0.11$, $p=.02$; $R_{\text{baseline}}=0.11$, $p=.10$) though the correlations are similar in magnitude (Fig 1). This pattern is not surprising given that net household income increased over time naturally (in part due to COVID-19-related social safety net expansions), thereby resulting in a broader distribution of income in the BFY sample at wave 4 than at study entry.

256 Impact of the Monthly Cash Gift on Maternal Bandwidth

257 We find no differences in the maternal MEFS score between the high- and low-cash gift
258 groups (Table 1). BFY mothers in the high-cash gift group had an average score of 97.59
259 ($SD=14.68$), while mothers in the low-cash gift group had an average score of 97.62 ($SD=14.77$).
260 We achieved sufficient statistical power to detect effect sizes of 0.2 SD or greater; we therefore
261 do not believe a lack of statistical power explains this lack of significant impact.

262 Due to our pre-registered directional hypothesis (i.e., maternal MEFS scores would be
263 higher in the high-cash gift group), we conducted a post hoc exploratory one-tailed test, which
264 also shows no significant impact of the high cash gift on maternal MEFS ($p=.32$).

265 **Table 1. Intent-to-treat effects of cash gift on maternal bandwidth at age 4.**

	Age 1 (2019–2020)		Age 2 (2020–2021)		Age 3 (2021–2022)		Age 4 (2022–2023)		Low- cash gift group mean	Pooled Ages 1–4 (2019–2023)	
	Estimate	<i>p</i> - value	Estimate	<i>p</i> - value	Estimate	<i>p</i> - value	Estimate	<i>p</i> - value		Estimate	<i>p</i> -value
MEFS standard score							0.552 [–1.425, 2.529]	.588	97.662	N/A	
Trouble concentrating (scale 0–4)	0.050 [–0.047, 0.148]	.329	0.037 [–0.054, 0.128]	.447	0.030 [–0.060, 0.121]	.524	0.023 [–0.083, 0.129]	.667	0.318	0.036 [–0.027, 0.098]	.117
Have any trouble concentrating	0.010 [–0.046, 0.067]	.724	0.024 [–0.027, 0.075]	.356	0.038 [–0.016, 0.091]	.173	0.035 [–0.023, 0.094]	.235	0.206	0.027 [–0.009, 0.063]	.054

266 *Note.* 95% confidence intervals in brackets, corresponding to unadjusted *p*-values. Effect sizes and *p*-values associated with gift group coefficient in regression with covariates.

267 Includes baseline covariates and child age at interview (in months).

We also find no impacts on the survey item regarding ability to concentrate (either as a continuous or dichotomized variable) at any survey wave nor pooled across four survey waves (see Table 1). This may be explained by limited variability in this measure: Most mothers reported no difficulty concentrating: 73% responded “Not at all.” Pooling across all four waves of data collection decreases the uncertainty of our intent-to-treat estimate on the concentration item (0.027 [−0.009, 0.063]), though this effect is not statistically significant ($p=.054$) and is in the opposite of the hypothesized direction (i.e., numerically more trouble concentrating in the high-cash gift group).

Timing of Cash Gift Effects on Maternal Bandwidth

We find no associations between the timing of MEFS assessment by week or by day and maternal MEFS scores. Descriptively, maternal MEFS scores were highest in the first week of the month and in the week of receipt of the cash gift, but these did not statistically differ from the third or fourth week of the month or the week prior to receipt of the cash gift. Interactions between the high-cash gift treatment and the day, week, or relative week from the cash gift monthly disbursement day are not statistically significant (Table 2, Fig 2, Fig 3).

Fig 2. MEFS standard scores by week. Number of weeks since cash gift disbursement (top row) and week of month (bottom row). The low-cash gift group is shown in the left panel; the high-cash gift group is shown in the right panel. Colored points represent means.

Fig 3. MEFS standard scores by day of assessment. Number of days since cash gift disbursement (top row) and day of month (bottom row). The low-cash gift group is shown in the left panel; the high-cash gift group is shown in the right panel. Colored lines represent loess curves for visualization purposes only.

290 An analysis of timing effects in the low-cash gift group only for comparison also showed
291 no statistical relationship between timing of MEFS assessment and maternal MEFS scores.
292 Additionally, we conducted analyses including interactions with site to account for potential site
293 differences in the timing of benefits, bill payments, etc.; these show no consistent statistical
294 interactions. Finally, we conducted analyses testing for a 3-way interaction between treatment
295 group, proximity to cash gift disbursement (in days), and before/after cash gift disbursement for
296 each measure of bandwidth; no statistically significant interactions were found.

297 **Table 2. Timing of cash gift receipt associations with maternal MEFS.**

	Model 1: Week of month		Model 2: Weeks since gift		Model 3: Day of month		Model 4: Days since gift	
Term	Estimate	<i>p</i> -value	Estimate	<i>p</i> -value	Estimate	<i>p</i> -value	Estimate	<i>p</i> -value
Intercept	104.72	<.001	105.25	<.001	105.67	<.001	104.67	.001
High-cash gift	0.20	.918	1.04	.597	0.15	.939	-0.38	.846
2nd week of month	-0.54	.767						
3rd week of month	-2.80	.138						
4th week of month	-1.76	.349						
High-cash gift × 2nd week of month	-0.99	.719						
High-cash gift × 3rd week of month	3.78	.184						
High-cash gift × 4th week of month	-2.26	.422						
1 week since previous gift payment			-0.49	.786				
2 weeks since previous gift payment			-2.22	.266				
3 weeks since previous gift payment			-1.50	.376				
High-cash gift × 1 week since previous gift payment			-2.52	.361				
High-cash gift × 2 weeks since previous gift payment			0.19	.948				
High-cash gift × 3 weeks since			0.28	.919				

	Model 1: Week of month		Model 2: Weeks since gift		Model 3: Day of month		Model 4: Days since gift	
Term	Estimate	<i>p</i> -value	Estimate	<i>p</i> -value	Estimate	<i>p</i> -value	Estimate	<i>p</i> -value
previous gift payment								
Day					−0.10	.182		
High-cash gift × Day					0.01	.938		
Days since previous gift payment							−0.05	.451
High-cash gift × Days since previous gift payment							0.05	.682

298 *Note.* Includes covariates for month of assessment, site, language of assessment, and baseline characteristics.

Discussion

This study reports an analysis of maternal cognitive bandwidth measured after four years of receipt of a monthly unconditional cash transfer in the ongoing BFY study. The study makes two main contributions to the literature. First, we provide estimates of the impact of monthly unconditional income on objective assessments of maternal bandwidth in a large sample of racially and ethnically diverse mothers with low income in the U.S.; our results rule out effect sizes of 0.2 SD or larger. This study provides new evidence of the effects of a cash transfer intervention given to parents in the context of a high-income country, thus testing one potential channel of how family resources might matter during a sensitive period of children's development.

Second, we conducted exploratory analyses examining whether within-month timing of the monthly cash gift, which is independent of other sources of income, has effects on maternal cognitive bandwidth. We find no impacts of the monthly unconditional cash gift on preregistered analyses of mothers' cognitive bandwidth, and no effects of the timing of the cash gift, nor interactions of that timing with the high-cash gift, throughout the month. We now turn to consider several possible factors that may have contributed to the lack of significant impact of the cash transfers on maternal cognitive bandwidth.

No Evidence for Impact of Cash Gift on Cognitive Bandwidth

Results of earlier waves of the study suggest that the cash gift increased household income and reduced poverty in this sample [30], though so far there are no improvements in maternal subjective well-being, maternal psychological health, or perceived or parenting stress (Magnuson et al., 2024, *under review*). On the one hand, it is possible that income does not exert a causal impact on bandwidth. The lack of significant impacts on maternal bandwidth aligns with

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findings from some prior studies [11–13]. On the other hand, there are several possible explanations for why there may be effects of poverty reduction on maternal bandwidth, but we do not detect them in the present study.

First, the lack of significant impacts may be due to broader context shifts that co-occurred over the 4-year duration of the cash gifts in the present analysis. For example, many families' income increased over this time period, perhaps lessening the effect of the \$313/month income supplement. Additionally, families across the United States experienced increased stress due to the COVID-19 pandemic, as well as increased uncertainty about the broader economy. Pandemic-related expansions of social safety net programs in the United States also increased household incomes for this sample, while inflation reduced the value of the cash gift. At the same time, the predictability of the cash gift may have decreased some of the income uncertainty faced by families during this time. These widespread contextual changes and countervailing influences may have attenuated the impact of the BFY cash gift intervention on maternal bandwidth.

Another possibility is that the cash gift may not have been large enough to generate psychological benefits on attention and effortful control as posited. The BFY cash gift significantly increased net income and significantly reduced poverty, yet more than 94% of families in this sample remain at or below 200% of the federal poverty threshold [30]. There is no evidence that the cash gift has reduced mothers' economic pressure, psychological distress, parenting stress, subjective well-being, hope, or agency [30,38]. This may suggest that the unconditional cash gifts in the present study were insufficient to alleviate the psychological effects of poverty for these mothers, consistent with findings from some previous studies [6]. In this study, we found that cognitive bandwidth was modestly but significantly correlated with household income in the previous calendar year, despite the limited range of income in our

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sample. This leaves open the possibility that significant differences in bandwidth may be apparent with a larger income range or a larger cash transfer.

Relatedly, it is possible that the cash gifts had some psychological effects for mothers without increasing their bandwidth overall. For example, mothers in the high-cash gift group devoted more time to their children (i.e., increased time spent in early learning activities; Gennetian et al., 2024), without cutting back on their time at work [39]. In this way, it may be that the cash gift allowed mothers to shift their attention to different inputs but did not reduce demands on their attention overall, although the present data cannot speak to this possibility.

A third possibility is that the MEFS and one item on ability to concentrate are not sensitive measures of maternal cognitive bandwidth. For example, the MEFS may capture broader, more stable neurocognitive executive functioning skills and may be insufficiently sensitive to capture shifts in more short-lived attentional demands. Financial scarcity may reduce bandwidth by leading people to shift their attention towards momentary scarcity-related demands at the expense of other demands [4,5], and such a shift may not be captured by these measures. Providing support to the interpretation that our two measures did not sufficiently capture variation in bandwidth in this sample, we find that our two measures of bandwidth are significantly but weakly correlated with each other ($r=.07$, $p=.04$). A future wave of data collection, when focal children are 6 years of age, will include MEFS and the concentration item as measured here, as well as a scarcity prime immediately preceding the Flanker task, a cognitive bandwidth measure used in previous research (Carvalho et al., 2016; Ong et al., 2019).

Limitations

This study has many strengths, as well as a few limitations, related to the direct assessment of maternal cognitive bandwidth within a randomized controlled study of a poverty

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reduction intervention in the U.S. Limitations include sensitivity of our measures to shifts in cognitive bandwidth, the fact that the study did not capture momentary financial scarcity or perceptions of financial stress, and broader economic and public health shifts (including those related to COVID-19) over the time period of analyses. Regarding our timing analyses, mothers had some control over the scheduling of the lab visit, so while timing of cash gift receipt throughout the month was exogenous, timing of assessment was partly endogenous.

Conclusion

Parenting involves a variety of demands on cognitive bandwidth and requires complex decision-making. For parents of young children, the cognitive consequences of poverty may have ripple effects on a variety of parenting behaviors and family processes. This study offers new empirical evidence about the causal impact of unconditional cash gifts on maternal cognitive bandwidth. We find no evidence that a monthly unconditional cash transfer provided for four years to mothers with low income, starting at the time of childbirth, improved their cognitive bandwidth. Future work will address open questions about continued cash receipt over time and more refined measurement of maternal attention.

Acknowledgements

Research reported in this publication was supported by the Eunice Kennedy Shriver National Institute of Child Health and Human Development of the National Institutes of Health under Award Number R01HD087384. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. This research was additionally supported by the US Department of Health and Human Services, Administration for Children and Families, Office of Planning, Research and Evaluation; Office

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391 of Behavioral and Social Sciences Research-Office of the Director, National Institutes Of Health;
392 Andrew and Julie Klingenstein Family Fund; Annie E. Casey Foundation; Arnold Ventures;
393 Arrow Impact; BCBS of Louisiana Foundation; Bezos Family Foundation; Bill and Melinda
394 Gates Foundation; Bill Hammack and Janice Parmelee; Brady Education Fund; Chan Zuckerberg
395 Initiative (Silicon Valley Community Foundation); Charles and Lynn Schusterman Family
396 Philanthropies; Child Welfare Fund; Esther A. and Joseph Klingenstein Fund; Ford Foundation;
397 Greater New Orleans Foundation; Heising-Simons Foundation; Holland Foundation; Jacobs
398 Foundation; JPB Foundation; J-PAL North America; Lozier Foundation; New York City Mayor's
399 Office for Economic Opportunity; Perigee Fund; Robert Wood Johnson Foundation; Robin
400 Hood; Sherwood Foundation; Valhalla Foundation; Weitz Family Foundation; W.K. Kellogg
401 Foundation; and three anonymous donors. Funding for S.E. additionally provided by the Abdul
402 Latif Jameel Poverty Action Lab. We thank the University of Michigan Survey Research Center,
403 our partners in recruitment, data collection, and participant location and retention; and the
404 participants of the Psychology and Economics of Poverty Convening 2024 for their feedback and
405 discussion.

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Supporting Information

518 **Fig A.1. CONSORT diagram for Baby's First Years maternal bandwidth measures**
519 **through age 4 data collection.**

520 **Table A.1. Baseline balance by high-cash and low-cash gift groups for n=828, wave 4**
521 **maternal MEFS analytic sample.**